1 Lecture 12 OLS in Practice Short Answer Questions

2 Question #1

Suppose that the one-way ticket price, $fare_i$, for airline ticket i is related to distance, measured in hundreds of miles, $dist_i$, by

$$\log(fare_i) = 4.666 + 0.043\, dist_i.$$

a. Write $fare_i$ as a function of $dist_i$ rather than $\log(fare_i)$.

b. What is the marginal effect of $dist_i$ on $fare_i$?

$$\frac{\partial fare_i}{\partial dist_i}$$

c. Let

$$100\frac{\partial fare_i}{fare_i}$$

approximate the percentage change in $fare_i$. Show that the coefficient 0.043 implies that increasing distance by 1 mile, rather than increasing $dist_i$ by one unit (which corresponds to 100 miles), is associated with an approximate 0.043% increase in fare.

3 Question #2

Suppose that the one-way ticket price, $fare_i$, for airline ticket i is related to distance, measured in hundreds of miles, $dist_i$, by

$$fare_i = 41 + 66\log(dist_i)$$

a. What is the marginal effect of $dist_i$ on $fare_i$?

$$\frac{\partial fare_i}{\partial dist_i}$$

b. Let

$$100\frac{\partial dist_i}{dist_i}$$

approximate the percentage change in $dist_i$. Re-write the marginal effect to indicate that the formula above shows that a 1% increase in $dist_i$ corresponds to a \$0.66 increase in $fare_i$.

c. Show that the interpretation of the coefficient 66 does not depend on the units of $dist_i$. For example, if $dist_i$ were measured in miles instead of hundreds of miles, show that a 1% increase in distance would still increase $fare_i$ by approximately \$0.66.

4 Question #3

Suppose that the one-way ticket price, $fare_i$, for airline ticket i is related to distance, measured in hundreds of miles, $dist_i$, by

$$\log(fare_i) = 4.25 + 0.40 \log(dist_i).$$

a. Write $fare_i$ as a function of $dist_i$ rather than $\log(fare_i)$.

b. What is the marginal effect of $dist_i$ on $fare_i$?

$$\frac{\partial fare_i}{\partial dist_i}$$

c. Let

$$100\frac{\partial fare_i}{fare_i} \qquad \text{and} \qquad 100\frac{\partial dist_i}{dist_i}$$

approximate the percentage changes in $fare_i$ and $dist_i$, respectively. Show that the coefficient 0.40 implies that a 10% increase in distance is associated with an approximate 4% increase in fare.